60 days Generative AI

AI
Artificial Intelligence
🤖 Any system that makes computers think and solve problems like humans

AI is the **biggest umbrella**. If a computer does something that normally needs human intelligence — playing a game, recognising your face, routing traffic — that's AI. ML, Deep Learning, GenAI are all types of AI.

Think of it as the **whole category**, not one specific technology.

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AI is the entire school building. ML, Deep Learning, GenAI are just different classrooms inside. You can't be in a classroom without being in the school.

👇 See Trending Examples Explained

▾

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Smart Traffic Systems (Delhi, Bengaluru)
TRENDING INDIA

Indian cities are now deploying AI traffic management — cameras at intersections monitor vehicle flow and **automatically adjust signal timings in real time.** If one road is jammed, the system extends the green light for that lane. Pure rule-based AI logic: "If density on Road A > threshold → give Road A more time." No learning involved — just smart, fast decisions.

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AI NPCs in Games (GTA VI, Elden Ring)
GAMING 2025

Non-player characters in modern games follow thousands of behaviour rules: **"If player enters zone → alert nearby guards. If health < 20% → retreat."** GTA VI is expected to have the most realistic NPC AI ever — pedestrians react to weather, time of day, and player behaviour using layered rule trees. This is classic AI logic running at incredible speed.

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UPI Fraud Rules Engine (NPCI)
INDIA FINTECH

Every UPI payment in India passes through NPCI's rule engine — **over 500 crore transactions per month.** Rules like "If same account receives 10+ payments in 60 seconds → flag" run instantly for every transaction. This rule-based AI layer is the first line of defence before ML models even analyse it.

ML
Machine Learning
📊 AI that learns patterns from examples — no manual rules needed

Instead of writing rules, you **show the machine thousands of examples** and it figures out the pattern itself. The machine finds connections in data that humans would never spot manually.

More data = smarter model. It keeps improving as it sees more examples.

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Like a child learning what "spam" looks like. Show them 10,000 spam messages — they start recognising patterns themselves. Next new spam? Caught instantly.

👇 See Trending 2025 Examples Explained
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Instagram Reels Feed Algorithm
TRENDING 2025

Every time you watch a Reel fully, skip one, or comment — Instagram's ML model updates your profile. It has learned patterns across **2 billion users**: "People who like cooking Reels also tend to like travel content." Your feed is rebuilt every few minutes based on your latest signals. **No human decides what you see — the ML model does, every single refresh.**

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Practo / Apollo AI Health Predictions
INDIA HEALTH TECH

Indian health platforms now use ML trained on millions of patient records. You enter symptoms → **ML model predicts probable conditions, suggests specialist type, and flags urgency level.** It learned from patterns like "fever + cough + fatigue in monsoon season = likely dengue in Mumbai zone." Gets smarter every monsoon season.

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Meesho / Flipkart Product Recommendations
INDIA E-COMMERCE

Meesho's ML analyses what Tier-2 and Tier-3 city shoppers browse, add to cart, and return. It found patterns: **"Women in UP who bought kurtas under ₹500 also tend to buy matching dupattas within 48 hours."** The "You might like" section is entirely ML-generated, boosting average order value by showing the right product at the right time.

Deep Learning
Neural Networks

🧠 ML with brain-like layers — understands images, voice and video on its own

Give it raw images, audio or video and it **figures out all the features itself — layer by layer**, just like your brain does. No human needs to point out what to look for. The more layers, the deeper the understanding.

It powers everything you see and hear in modern apps.

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Like peeling an onion. Layer 1 sees edges → Layer 2 sees shapes → Deeper layers understand faces, objects, words. Each layer builds on the last.

👇 See Trending 2025 Examples Explained

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Snapchat / Instagram AI Filters
USED DAILY

Those dog ears, beauty filters, and face-swap effects? That's deep learning running live on your phone. **A deep CNN finds 68 facial landmarks — eyes, nose, jawline, lips — in real time at 30 frames per second.** It tracks your face as you move, tilts, smiles, and places virtual objects perfectly on your face. All of this happens in under 10 milliseconds per frame.

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Google Lens / Search by Image
TRENDING 2025

Point your camera at anything — a plant, a dress, a restaurant menu in Hindi — and Google Lens identifies it instantly. **Deep learning analyses visual features layer by layer:** shape, texture, colour, context. It was trained on billions of labelled images. Now it can identify 10,000+ plant species, read handwritten text, and even translate menus live through your camera.

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Whisper / Real-Time AI Transcription
2024–25 HOT TOOL

OpenAI's Whisper and tools like Otter.ai use deep learning to transcribe meetings live — **including Indian accents, Hinglish mixing, and background noise.** A deep neural network converts sound waves into text, understanding context to fix errors automatically. Journalists, YouTubers, and startups use this daily to get auto-generated subtitles and meeting notes.

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Deepfake Detection Tools (2025)
CRITICAL RIGHT NOW

With deepfakes flooding social media, platforms like Meta and startups like Hive now use deep learning to detect AI-generated fake videos. **The model analyses micro-expressions, blinking patterns, and pixel inconsistencies around the hairline** — things invisible to humans. It was trained on millions of real and fake videos. India's Election Commission used similar tools in the 2024 elections.

Generative AI
GenAI · LLMs · The ChatGPT Era

✨ AI that creates brand new things — text, images, video, music, code from a prompt

All previous AI understood existing things. Generative AI **creates things that never existed before.** Trained on nearly the entire internet, it composes original text, images, audio and video from a single instruction.

This is what changed everything in 2022 — and it's still exploding in 2025.

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Like someone who has read every book ever written. Ask them to write a new story — they don't copy. They create something original from everything they've absorbed.

👇 See Trending 2025 Examples Explained

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Claude 3.5 / GPT-4o — Everyday AI Assistant

MOST USED 2025

Millions of people now use Claude or ChatGPT as their **daily thinking partner** — drafting emails, explaining complex topics, writing code, preparing for interviews, summarising long documents. When you type "Explain GST in simple Hindi" — the model generates a fresh, tailored explanation word-by-word that **has never been written before in exactly that way.** It's not searching the internet — it's composing.

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Kling AI / Sora — Text to Video

🔥 HOTTEST 2025

Type: **"A street food vendor making pav bhaji in Mumbai rain, cinematic, slow motion."** Kling AI generates a photorealistic 30-second video in minutes — no camera, no crew, no editing software. Indian content creators and ad agencies are already using this to make ads at 1/100th of traditional cost. **Bollywood production houses are testing it for pre-visualisation of scenes.**

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Suno v4 — AI Music Generation

VIRAL 2025

Type: **"Bhojpuri folk song about monsoon harvest, male vocals, dhol and flute."** Suno v4 generates a fully produced song in 30 seconds — real-sounding vocals, instruments, mixing, mastering. Indian regional music creators are going viral on YouTube by generating songs in Bhojpuri, Rajasthani, and Marathi using AI. **Zero music production skills needed.**

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Cursor AI / GitHub Copilot — AI Code Editor

DEV TOOL OF 2025

Cursor is the fastest-growing coding tool of 2025. You describe what you want: **"Build a login page with OTP verification using React"** — and Cursor writes the complete code, explains each part, fixes bugs as you go. Indian developers report finishing in **1 day what used to take a week.** It's not just autocomplete — it understands your entire codebase.

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Midjourney v7 / Adobe Firefly — AI Image Creation

CREATIVE INDUSTRY

Indian graphic designers and wedding photographers are using AI image generators for **concept art, social media posts, invitation card designs, and advertising mockups.** A designer types "Traditional Rajasthani bride, golden hour, magazine cover style" and gets 4 stunning images in 10 seconds. What took 4 hours of Photoshop now takes a single prompt. **Agencies are creating entire ad campaigns with AI visuals.**

Agentic AI
AI Agents · Hottest Topic 2025
🔥 AI that doesn't just answer — it plans steps, uses tools, and completes real tasks for you

GenAI answers inside a chat box. Agentic AI **steps out and does things in the real world.** Give it a goal — not a question — and it figures out all the steps, opens apps, browses websites, writes code, sends emails, and delivers the finished result.

Like a real digital employee — not just a chatbot.

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GenAI = Smart advisor who tells you what to do.
Agentic AI = Smart employee who actually does it — plans, executes, delivers, reports back. You just give the goal.

👇 See Trending 2025 Examples Explained

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Claude Cowork — AI Desktop Agent
ANTHROPIC 2025

You say: **"Sort all my downloaded invoices by month, rename them properly, and make a summary Excel sheet."** Claude Cowork opens your file manager, reads every PDF, creates folders like "Jan_2025_Invoices", renames files, opens Excel, builds the summary table with totals — and notifies you when done. **You gave one sentence. Claude did 20 steps across 3 applications.** You never touched your keyboard after that.

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Gemini Deep Research / OpenAI Deep Research
🔥 MOST VIRAL 2025

You ask: **"Give me a full analysis of the Indian EV market — top players, funding, government policies, growth numbers."** The AI agent spends 20 minutes autonomously — searches 60+ websites, reads full articles, cross-checks facts, discards outdated sources, and writes a structured 10-page report with data and citations. **What takes a research analyst 2 full days takes the agent under 30 minutes.** Consultants and students are using this every day.

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Devin 2.0 — AI Software Engineer
DEV WORLD 2025

A startup posts a task: **"Add dark mode to our website."** Devin — an AI agent — reads the entire codebase, understands the design system, writes the CSS and JavaScript changes, runs automated tests, fixes errors it finds, and opens a GitHub Pull Request for human review. **A task that takes a junior developer 3 hours is done by Devin in 12 minutes.** Indian IT firms are already testing AI agents for bug fixing tasks.

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Zapier AI Agent / Make.com Automations
BUSINESS TOOL 2025

A small business owner sets up an AI agent: **"When a new lead fills our website form → check if they match our ideal customer profile → if yes, send personalised intro email → add to CRM → schedule a follow-up reminder for Day 3."** The agent runs 24/7 — it reads the form, analyses the lead, writes a personalised email (not a template), and updates 3 different tools. **One person now handles the work of a 3-person sales team.**

🌐

Perplexity AI — Answer + Action Agent
REPLACING GOOGLE

Perplexity is the fastest-growing AI tool of 2025 — adding **1 million new users every week.** Unlike Google which shows links, Perplexity's agent actually reads those pages, synthesises the answer, cites sources, and can take follow-up actions like booking, comparing prices, or drafting a response. Students in India are using it to research assignments, and professionals use it to prep for client meetings — **getting a researched answer in 10 seconds instead of spending 30 minutes on Google.**

AI → Computer that think like humans

ML → AI that learns from example automatically — finds patterns without manual rules.

DL → Brain-like AI with many layers — understanding image, voice, or words.

GenAI → Creates brand new content — text, videos, music, code from single prompt

Agentic AI → Plan steps & take real action.

Day 2 — Discriminative vs Generative Models

Both Discriminative and Generative models are **subsets of Machine Learning (ML) and Deep Learning**, which is itself a subset of **Artificial Intelligence (AI)**.

🎯 **DISCRIMINATIVE MODELS — "The Judge"**

What it?

A discriminative model looks at data and decides which category it belongs to. That's it. It doesn't create anything. It doesn't understand what data looks like from the inside. It just draws an invisible boundary line between groups and uses that line to judge new inputs.

Think of it like this — a doctor looks at your symptoms and says "this is diabetes" or "this is not diabetes." He's not creating a new patient from scratch. He's just judging based on what he sees.

Math behind it: It learns P(Y|X) — given this input X, what is the probability of label Y?

Analogy: SBI loan officer — looks at your CIBIL score and says Approve or Reject.

Types:

- **Logistic Regression** — yes/no output. Example: spam or not spam
- **SVM** — finds best boundary between classes. Example: tumour or no tumour
- **Decision Tree / Random Forest** — yes/no questions in a tree. Example: CIBIL credit scoring
- **CNN (classifier)** — classifies images. Example: DigiYatra face match
- **RNN / LSTM** — for sequences. Example: stock market direction prediction
- **BERT** — reads text both ways, classifies it. Example: IRCTC complaint categorization

✨ GENERATIVE MODELS — “The Creator”

What is it?

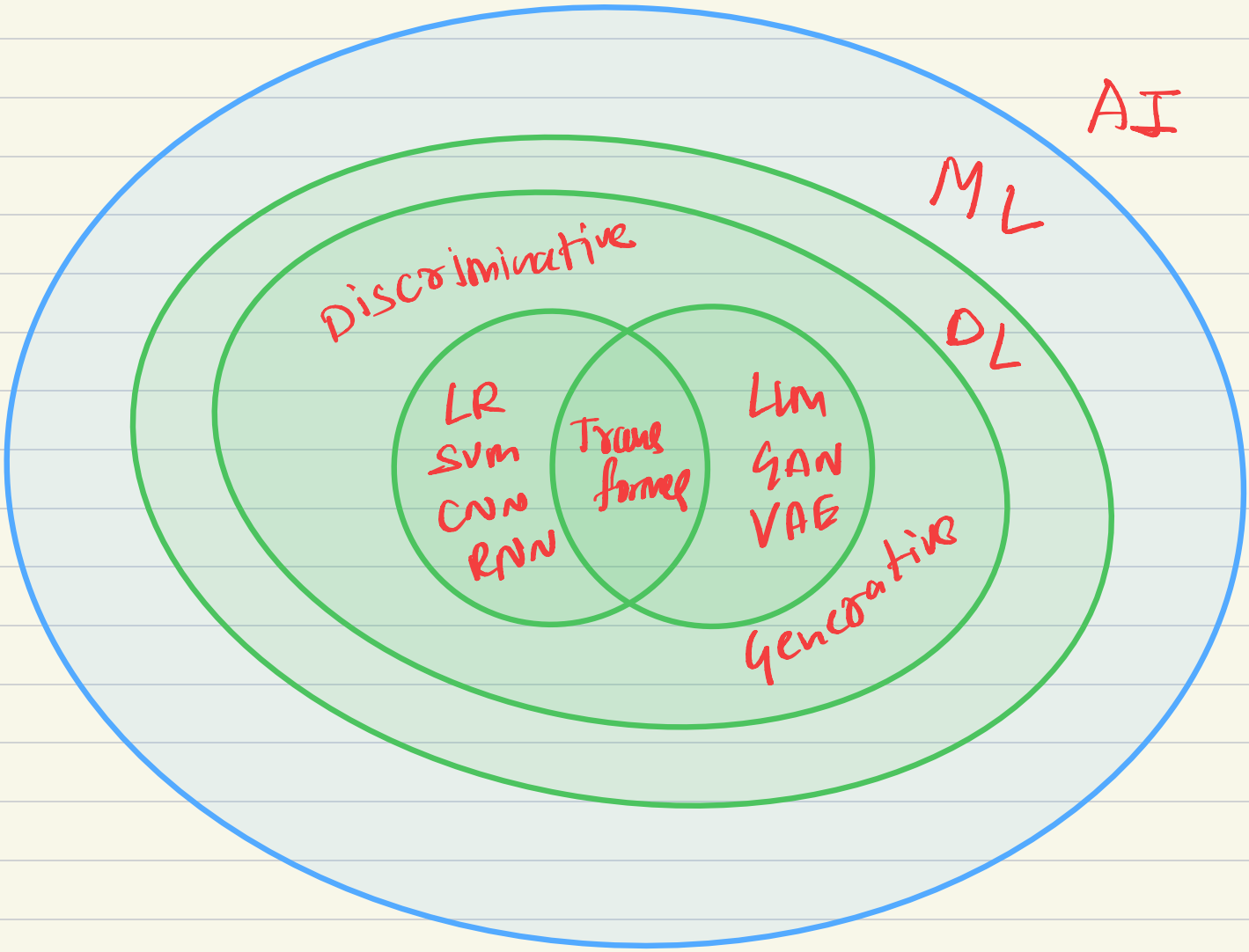
A generative model doesn’t just judge data — it learns what data looks like from the inside so deeply that it can create brand new data that has never existed before but feels completely real.

Think of it like this — A.R. Rahman has absorbed 10,000 songs over his lifetime — classical, folk, Bollywood. He understands music so deeply from the inside that he can now compose a completely new song that no one has ever heard — but it sounds authentically Indian. He’s not classifying music genres. He’s creating music. That is generative thinking.

Math behind it: It learns P(X) — the full probability distribution of the data itself — not just the boundary between classes.

One line definition: “What would new data in this world look like?”

Types:

1. **LLMs** — predict next word, again and again → generates text
Examples: Claude Sonnet 4.6, GPT-4o, Gemini 2.5 Pro, LLaMA 3.3, Krutrim (India)
2. **GANs** — Generator creates fakes, Discriminator catches fakes, they train each other
Examples: StyleGAN3. Used for product photo generation (Meesho/Myntra)
3. **VAEs** — compresses data into latent space, samples from it, generates new data
Examples: Used inside Stable Diffusion’s image compression layer
4. **Diffusion Models** — adds noise to image until it’s static, then learns to reverse it
Examples: Flux.1, DALL-E 3, Midjourney v6, Stable Diffusion 3.5, Sora (video)
5. **Flow-based Models** — transforms data into noise through reversible steps, then reverses to generate
Examples: WaveGlow — used in Sarvam.ai for Hindi/Tamil voice synthesis
6. **Autoregressive Image Models** — treats image patches like words, generates patch by patch
Examples: Chameleon (Meta 2024), parts of GPT-4o

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ARTIFICIAL INTELLIGENCE
└── MACHINE LEARNING
    └── DEEP LEARNING
        ├── 🎯 DISCRIMINATIVE — "What IS this?"
        │   ├── Logistic Regression
        │   ├── SVM
        │   ├── Decision Tree / Random Forest
        │   ├── CNN (as classifier)
        │   ├── RNN / LSTM
        │   └── BERT (Encoder only)
        │
        └── ✨ GENERATIVE — "Create something NEW"
            ├── LLMs (Claude, GPT-4o, Gemini 2.5)
            ├── GANs (StyleGAN3)
            ├── VAEs
            ├── Diffusion Models (Flux.1, DALL-E 3, Sora)
            ├── Flow-based Models (WaveGlow)
            └── Autoregressive Image Models (Chameleon)
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